HAIYI MEI

Research Fellow

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TECH STACK

Python | C++ | Git | Unreal Engine | Blender | PyTorch | Docker

ACQUAINTED

Game Engine | PBR | Synthetic Data | SMPLX | SaaS | NeRF | Diffusion | Differentiable Rendering | Music (Piano/Guitar/Theory)

EDUCATION

Master's Degree

Shandong University

2019 – 2022

Visiting Research Intern

NLPR | CASIA

2018 – 2021

Bachelor's Degree

NJUST

2015 – 2019

- Head of the Symphony Orchestra of School

LANGUAGES

Chinese: Native

English: C1

ABOUT ME

Research Fellow at SenseTime Research, focusing on 3D synthetic data generation for computer vision tasks.

EXPERIENCE

Mid-Level Research Fellow | SenseTime

2021 – Present

- Spearheaded the development of XRFeitoria, a rendering toolbox that streamlines synthetic data generation. This tool simplifies the construction of rendering pipelines in diverse areas, enabling contribution to publishing 5 papers within 1 year.
- Mainrole in designing SynBody, a large-scale synthetic dataset with layered human models.
  - Participated in designing SMPL-XL, which enriches SMPL-X in hair, garments, accessories, and textures.
  - Built a System of synthetic data generation as a Software-as-a-Service (SaaS) platform, including 1) layered human creation, 2) motion retargeting, 3) scene composition, 4) rendering pipeline, and 5) flow control.
  - Published a paper presented at ICCV 2023 as co-first author.
- Proficient in crafting advanced technology demonstration videos.

OPEN-SOURCE PROJECTS

XRFeitoria | Synthetic Data Rendering Toolbox

- Control over engine (UE/Blender) through RPC using system python.
- Render photorealistic images with ground-truth annotations.
- Manage assets/cameras, including import, place, export, and delete.
- CLI tools to render images from a mesh file.
- Streamline building rendering pipelines across various domains, effectively implemented in over 6 projects.

AWARDS

Patents
Hold 4 patents filed with CNIPA
CN108453742B, CN113011334A, CN114429436A, CN115272548A

Music Award – 2017
First Place Award, National University Piano Competition
Chinese Golden Bell Award for Music